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Program 1: Build a Docker Container from a Custom Dockerfile

Project structure:

```
.
├── Dockerfile
├── index.html
├── package.json
├── package-lock.json
├── App.css
├── App.jsx
├── assets
│   └── react.svg
├── index.css
└── main.jsx
```

STEP 1: Create a Dockerfile and add the following content.

```
FROM node:latest
WORKDIR /app
COPY package*.json .
RUN npm install
COPY . .
EXPOSE 5173
CMD ["npm", "run", "dev", "--", "--host", "0.0.0.0"]
```

STEP 2: Execute the Docker Build command on the Terminal to build the Docker image and then execute the Docker run command specifying the Port Number for the container.

› **sudo docker build -t client .**

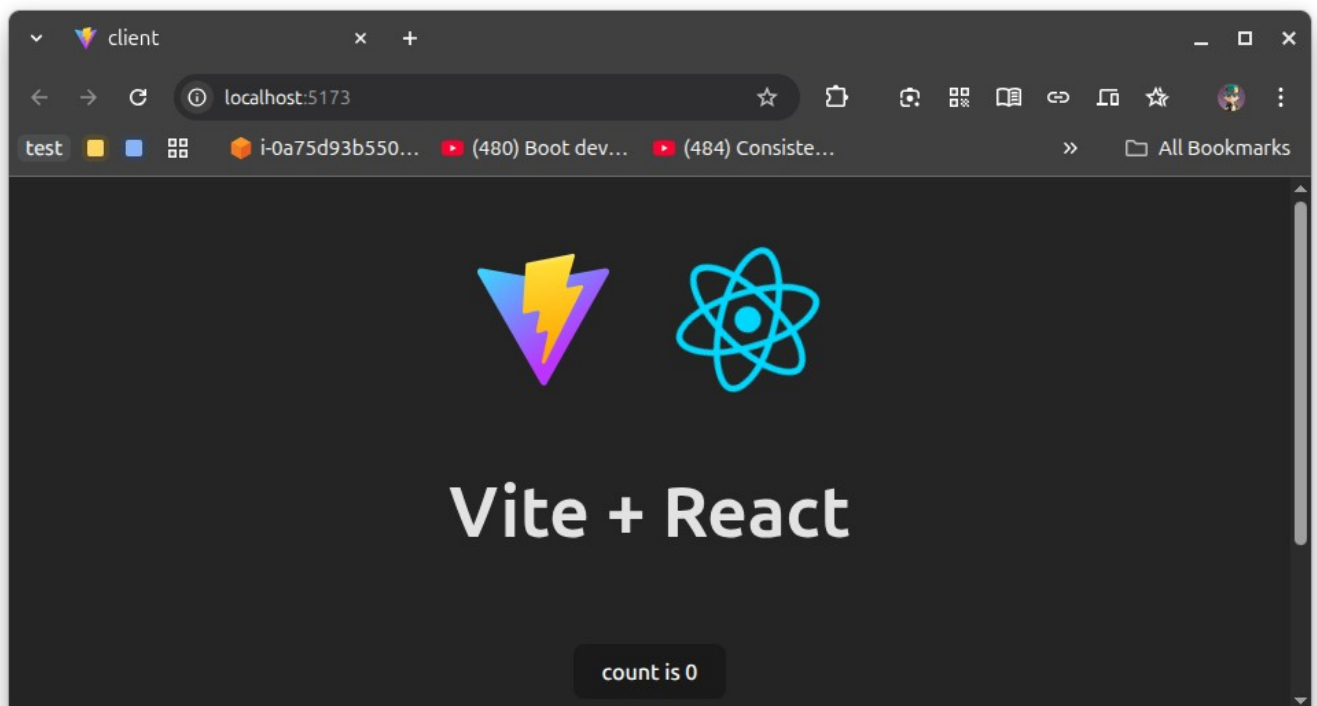
```
> sudo docker build -t client .
[+] Building 21.4s (4/9)                                docker:default
=> [internal] load build definition from Dockerfile      0.0s
=> => transferring dockerfile: 181B                      0.0s
=> [internal] load metadata for docker.io/library/node:latest 0.4s
=> [internal] load .dockerignore                        0.0s
=> => transferring context: 2B                             0.0s
=> [1/5] FROM docker.io/library/node:latest@sha256:0b1fad950f54a1d6f9e8e580205c157b4 21.0s
=> => resolve docker.io/library/node:latest@sha256:0b1fad950f54a1d6f9e8e580205c157b43 0.0s
=> => sha256:0b1fad950f54a1d6f9e8e580205c157b43315d2c4231c3a0b78137d8 5.14kB / 5.14kB 0.0s
=> => sha256:5d93aea697980315f27f81c68582d14f63dd3579c2d3a27dc495a 48.48MB / 48.48MB 14.0s
=> => sha256:bb445e472b1bad54f5a28edd51b11aec79eca8513394866a261891 5.24MB / 24.03MB 21.0s
=> => sha256:2123190679e81d983648da92f1bb9ddc74383512edb00ad64f93d 24.12MB / 64.40MB 21.0s
=> => sha256:260d5cf808df32b294d6eb34f31fef285b2e7fdc0986989517607b37 2.49kB / 2.49kB 0.0s
=> => sha256:5cdc8ed4f59b12fd00bf3cf863adeb51841e9c215f3d2b03842d77f7 6.75kB / 6.75kB 0.0s
=> => extracting sha256:5d93aea697980315f27f81c68582d14f63dd3579c2d3a27dc495a588279ed 0.7s
=> => sha256:32885a2b0a589e832bf6b250bd35a528b268360f166af2cd7094 19.92MB / 211.45MB 21.0s
=> [internal] load build context                        0.0s
=> => transferring context: 542B                          0.0s
```

➤ **sudo docker run -p 5173:5173 client**

```
> sudo docker run -p 5173:5173 client
> client@0.0.0 dev
> vite --host 0.0.0.0

VITE v7.1.12 ready in 135 ms
→ Local:   http://localhost:5173/
→ Network: http://172.18.0.2:5173/
```

STEP 3: visit the link to test if the docker Is perfectly running on the port 5173 on host with react app.



STEP 4: Stop and remove the container with the container number, remove the docker container and delete the Docker Image

➤ **sudo docker rm client1**

➤ **sudo docker images rm client**